# Inhwa Song • Statement of Purpose

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Many challenges at both individual and societal levels stem from the limitations of personal worldviews and perspectives. I have witnessed how such limitations can contribute to prejudice and division through serving as a student president during the COVID crisis and my two years as a developer and content moderator for the school's online community. At an individual level, rigid worldviews often result in struggles with mental health, knowledge I learned while training as a peer counselor. Together, these experiences have guided my interest to **harness the potential of social and language technologies to empower individuals to expand their worldviews, fostering psychological growth and societal harmony.** To this end, I hope to pursue my doctoral studies at the University of Washington.

My research interest has both guided and been shaped by my involvement in many exciting research projects across diverse academic domains and industry settings. During my exchange period at Georgia Tech, I led timely research under the mentorship of **Prof. Munmun De Choudhury** [2], to examine how individuals understand and use LLM chatbots when seeking support for their mental health needs in real-world settings. I conducted semi-structured interviews with 21 participants from every inhabited continent by connecting with local forums and support groups. Grounding analysis in psychotherapy literature of what makes support effective, we introduced the concept of therapeutic alignment. This work I led has been successfully submitted to *CSCW'24*, currently undergoing review following a request for revisions. I realized that providing interaction components for the user to gain control within a reliable structure is the key design consideration of building responsible generative AI systems, further motivating me to not only envision but to build and test the reliability of such systems.

This motivation led me to a research internship at NAVER AI Lab to work with **Dr. Young-Ho Kim**, whom I have long admired for his novel system contributions to HCI research. I developed ExploreSelf, an LLM-driven interactive system that supports users in exploring their narratives of personal challenges by actively engaging with adaptive guidance [1], grounding in a formative study. Even with my vast hands-on experience managing the development and deployment of complex computing systems, this particular work was especially valuable to me for three main reasons. First, I gained a deeper understanding of what system contribution means in HCI research. Second, this research I led received a *Revise & Resubmit decision in CHI'25* with compliments for the high research quality. Last but not least, I recognized my ability to quickly learn and apply new technologies and programming methods, along with the confidence that I can effectively tackle any new technical challenges I face in future research.

During my internship at NAVER AI Lab, I could communicate with AI researchers from diverse backgrounds, each bringing a different worldview. For instance, within the goal of responsible AI, stakeholders' perspectives diverged between creating safety benchmarks, embedding safety in LLMs, and investigating what responsible AI means in the context of Korean society. Through direct conversations with stakeholders, I saw firsthand how such divergent views stemmed from the tension between qualitative societal contexts and the quantitative methods that dominate ML. These discussions reinforced my commitment to promote the harmonical collaboration between diverse sectors with different worldviews. They also motivated me to broaden my own worldview on research. While I have translated qualitative insights into concrete system design [2], I also wanted to tackle research questions that involve direct intervention in language models. Under the guidance of **Prof. Alice Oh**, my ongoing project involves constructing a dialogue dataset of unstructured learner inputs paired with concise queries by fine-tuning the Phi-3 model with the StackExchange dataset, where each post combines a body of long-form explanations and a title of a focused question.

These research projects gave me the opportunity to adopt a wide range of approaches, from system building and evaluation [1,8], large-scale data analysis [3], fine-tuning language models, to qualitative methods such as semi-structured interviews [2] and cultural probe [5], and also systematic reviews [4]. Above all, I have developed my ability to identify valuable research questions and learned how to transform an initial idea into systematic research.

My vision is to support individuals to expand their worldviews by fostering psychological flexibility. This not only aims for personal growth but also to embrace pluralistic worldviews harmonically coexist. To this aim, I am excited to examine research questions at the intersection of HCI, NLP, and social computing for my PhD. For example, I am interested in developing tools to support challenging one's preconceptions. In real-world contexts, openly sharing personal prejudices or engaging with contrasting perspectives is often constrained by social norms and stigmas. While some turn to anonymous online platforms to safely express their thoughts, these spaces frequently fall victim to echo chambers, reinforcing biases rather than challenging them. Inspired by the Human Library[1], I propose leveraging AI agents as a powerful tool to address these challenges. These can serve as testbeds where individuals can safely explore their thoughts and experiment with engaging in dialogues that push the boundaries of their worldview. By providing opportunities for structured, judgment-free interactions, these systems could prepare users for real-world social interactions with greater understanding.

With my focus and potential directions to keep in mind, the University of Washington CSE would be a fantastic place to pursue my PhD. In particular, I aspire to receive the tutelage of **Prof. Tim Althoff**, whose research on human-AI collaboration for psychological and mental health support has inspired my first-authored work [1,2]. I hope to build on this direction by incorporating broader approaches and drawing on his insights in advancing and applying NLP to develop effective human-AI collaboration for psychological support. I am also keen to work with **Prof. Amy X. Zhang**, where her focus on designing and building social computing systems, involving multiple people (and agents) interacting together, will provide a rich context to my proposed research. My experience in developing and deploying a learnersourcing system for real-class use and serving as the developer of my university's online community will foster strong synergy in our collaboration.

Upon completing my PhD, I aspire to remain in academia as a professor, where I will strive to not only identify and solve socially critical research questions but also bridge the communication gap between diverse research sectors for real-world impact. With my rich research experience, commitment to broadening my worldview, and dedication to fostering social good, I am confident in my ability to succeed in the University of Washington CSE PhD program and contribute meaningfully to their mission of advancing research for societal impact.

References:

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